SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I **P Dinesh Reddy / 192525399** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Criteria	**Relevance to Application (1.25)**	**Use of Cloud Concepts (1.25)**	**Practical Insight (1)**	**Completeness (0.75)**	**Clarity (0.75)**	**TOTAL (5)**
Weightage	**25%**	**25%**	**20%**	**15%**	**15%**	**100%**
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty Total Marks Awarded:

Application Based Questions

1. How does cloud federation improve resource sharing and availability across datacenters?

Answer

Cloud Federation is the collaboration of multiple cloud providers or datacenters to share computing resources and services.

Resource Sharing

- Enables sharing of computing, storage, and network resources.
- Balances workloads across multiple datacenters.
- Reduces resource wastage.
- Supports dynamic allocation of resources based on demand.

Improved Availability

- If one datacenter fails, workloads are shifted to another.
- Ensures continuous service availability.
- Provides disaster recovery and backup support.
- Reduces service downtime.

Benefits

- High availability
- Better resource utilization
- Improved scalability
- Cost efficiency
- Increased reliability

Conclusion

Cloud federation improves resource sharing by allowing multiple datacenters to cooperate, resulting in higher availability, better performance, and efficient resource utilization.

2. What are the key security issues in virtualization, and how can they be mitigated?

Answer

Security Issues

1. **VM Escape**

- A malicious VM gains access to the hypervisor.

2. **Hypervisor Attacks**

- Attackers compromise the hypervisor and all hosted VMs.

3. **Data Leakage**

- Sensitive information is exposed due to weak isolation.

4. **Side-Channel Attacks**

- Attackers exploit shared hardware resources such as CPU cache.

5. **Unauthorized Access**

- Weak authentication allows attackers to access virtual resources.

Mitigation Techniques

- Keep hypervisors updated with security patches.
- Use strong VM isolation.
- Enable Multi-Factor Authentication (MFA).
- Encrypt data at rest and in transit.
- Apply Role-Based Access Control (RBAC).
- Deploy firewalls and Intrusion Detection Systems (IDS).
- Monitor systems continuously.

Conclusion

Proper security practices significantly reduce virtualization risks and improve cloud security.

3. How does a software-defined datacenter enhance scalability and management?

Answer

A **Software-Defined Datacenter (SDDC)** virtualizes compute, storage, and networking resources and manages them through software.

Scalability

- Resources can be added or removed dynamically.
- Supports automatic scaling based on workload.
- Improves workload distribution.

Management

- Centralized management of all infrastructure.
- Automation reduces manual configuration.
- Faster deployment of applications.
- Easier monitoring and maintenance.

Advantages

- Better resource utilization
- Reduced operational cost
- High flexibility
- Faster provisioning
- Improved efficiency

Conclusion

SDDC enhances scalability and simplifies infrastructure management through automation and centralized control.

4. How can identity and access control ensure data privacy in cloud-based healthcare systems?

Answer

Identity and Access Control ensure that only authorized users can access sensitive healthcare data.

Identity Management

- Assign unique identities to doctors, nurses, patients, and administrators.
- Verify user identities before granting access.

Access Control

- Role-Based Access Control (RBAC)
- Least-privilege principle
- Multi-Factor Authentication (MFA)
- Single Sign-On (SSO)

Data Privacy Measures

- Encrypt patient records.
- Maintain audit logs.
- Monitor user activities.
- Follow healthcare privacy regulations.
- Regularly review access permissions.

Benefits

- Protects patient confidentiality
- Prevents unauthorized access
- Ensures regulatory compliance
- Improves trust and accountability

Conclusion

Strong identity management and access control protect healthcare information while ensuring secure and authorized access.

5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Answer

Virtualization enables multiple virtual machines (VMs) to run on a single physical server, while a **hypervisor** manages these VMs.

Support for Scalability

- Quickly create or remove VMs.

- Scale applications based on demand.
- Enable load balancing.
- Support live migration of VMs.

Resource Efficiency

- Better utilization of CPU, memory, and storage.
- Reduce hardware requirements.
- Share physical resources among multiple VMs.
- Lower operational costs.

Role of Hypervisors

- Allocate resources to VMs.
- Isolate virtual machines.
- Monitor VM performance.
- Improve system reliability and availability.

Benefits

- High scalability
- Efficient resource utilization
- Cost savings
- Simplified management
- Improved flexibility

Conclusion

Virtualization and hypervisors enable cloud applications to scale efficiently while maximizing hardware utilization and reducing operational costs.